Polynomials – 9

Cyclotomic polynomials and the arithmetic of roots of unity

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Theoretical minimum. A root of unity is a rigid algebraic number: its minimal polynomial is cyclotomic, so every linear relation between roots of unity is a divisibility in $\mathbb{Z}[x]$. Over $\mathbb{F}_p$ the same rigidity appears in the power sums $\sum_x x^k$.

Definitions. The n-th *cyclotomic polynomial* $\Phi_n(x)$ is the product of $x - e^{2\pi ik/n}$ over $1 \le k \le n$ with $(k,n) = 1$; its roots are the *primitive* n-th roots of unity and $\deg \Phi_n = \varphi(n)$. A polynomial $f \in \mathbb{F}_p[x]$ is a *permutation polynomial* if $x \mapsto f(x)$ is a bijection of $\mathbb{F}_p$. A polynomial $P = \sum_j c_j X^j$ *annihilates* a function f (for a step h) if $\sum_j c_j f(x + jh) \equiv 0$, i.e. $P(T)f = 0$ for the shift $(Tf)(x) = f(x + h)$.

Theorem 1 (basic identities). $x^n - 1 = \prod_{d|n} \Phi_d(x)$, so $\Phi_n \in \mathbb{Z}[x]$ is monic (Trig-2 Roots of unity) and by Möbius inversion (Number theory – 6) $\Phi_n = \prod_{d|n} (x^d - 1)^{\mu(n/d)}$. Examples: $\Phi_p = 1 + x + \cdots + x^{p-1}$, $\Phi_{p^a}(x) = \Phi_p(x^{p^{a-1}})$ and $\Phi_{pq} = \frac{(x^{pq}-1)(x-1)}{(x^p-1)(x^q-1)}$ for primes $p \ne q$. Also Φ_n is the gcd of the polynomials $(x^n - 1)/(x^{n/p} - 1)$ over the primes $p \mid n$.

Theorem 2 (Gauss, Dedekind). Φ_n is irreducible over $\mathbb{Q}$: it is the minimal polynomial of each primitive n-th root of unity ζ, and $[\mathbb{Q}(\zeta) : \mathbb{Q}] = \varphi(n)$.

Proof idea. Let $f \in \mathbb{Z}[x]$ be the monic minimal polynomial of ζ (Polynomials – 5) and $\Phi_n = fg$; it suffices to show $f(\zeta^p) = 0$ for primes $p \nmid n$. Otherwise $g(\zeta^p) = 0$, so $f(x) \mid g(x^p) \equiv g(x)^p \pmod p$, hence $x^n - 1$ has a multiple factor over $\mathbb{F}_p$, which is false since $\gcd(x^n - 1, nx^{n-1}) = 1$ there.

Theorem 3 (vanishing sums). For $\zeta = e^{2\pi i/n}$ and $a_k \in \mathbb{Q}$ one has $\sum_{k<n} a_k \zeta^k = 0$ if and only if $\Phi_n \mid \sum_k a_k x^k$. For $n = p$ this means $a_0 = \cdots = a_{p-1}$; for $n = pq$ and $a_k \in \mathbb{Z}_{\ge 0}$ it means that the roots form a union of regular p-gons and q-gons centred at 0 (Problem 2). *Lam–Leung* (may be quoted): m roots of unity of order dividing n, repetitions allowed, can have zero sum iff $m \in p_1\mathbb{Z}_{\ge 0} + \cdots + p_r\mathbb{Z}_{\ge 0}$, where $p_1, \ldots, p_r$ are the primes dividing n.

Theorem 4 (functions on $\mathbb{F}_p$). $\sum_{x\in\mathbb{F}_p} x^k$ equals -1 if $k>0$ and $(p-1)\mid k$, and 0 otherwise ($0^0=1$): sum a geometric progression in a generator of $\mathbb{F}_p^*$ (Number theory – 2). Every map $\mathbb{F}_p\to\mathbb{F}_p$ is a unique polynomial of degree $\le p-1$ (Polynomials – 6), and $\sum_x f(x)$ is minus its coefficient at x^{p-1}. *Hermite's criterion:* f is a permutation polynomial iff f has exactly one root in $\mathbb{F}_p$ and for each $1\le t\le p-2$ the remainder of f^t modulo x^p-x has degree $\le p-2$. *Idea:* the number of solutions of $f(x)=a$ is $\equiv\sum_x\bigl(1-(f(x)-a)^{p-1}\bigr) \pmod p$.

Fact 1 (annihilating ideals). The polynomials annihilating f form an ideal (Q) of $\mathbb{C}[X]$, closed under gcd; a sequence satisfies a linear recurrence iff $Q\neq 0$ (Combinatorics – 2). Typical moves: find two annihilators with a small gcd, or show the ideal is stable under $P(X)\mapsto P(X^m)$, so the roots of Q are closed under $\lambda\mapsto\lambda^m$.

Fact 2 (traces). If α is algebraic of degree d with conjugates $\alpha_1,\dots,\alpha_d$, its *trace* $\operatorname{Tr}\alpha=\alpha_1+\cdots+\alpha_d$ is rational (Vieta), and an integer if α is an algebraic integer. For $\gamma\in\mathbb{Q}(\zeta)$, $\zeta=e^{2\pi i/n}$, the sum of $\sigma_s(\gamma)$ over the automorphisms $\sigma_s\colon\zeta\mapsto\zeta^s$, $\gcd(s,n)=1$, equals $\frac{\varphi(n)}{\deg\gamma}\operatorname{Tr}\gamma$; it is $\mathbb{Q}$-linear in γ. For example the sum of the primitive n-th roots of unity is $\mu(n)$, and $\sum_s\sigma_s(r)=\varphi(n)r$ for rational r.

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1. Prove Theorem 2 in full, including the passage from primes $p\nmid n$ to all exponents coprime to n. Then prove that $\Phi_n(1)=p$ if $n=p^a$ with $a\ge 1$ and $\Phi_n(1)=1$ if $n>1$ is not a prime power, and that the sum of the primitive n-th roots of unity equals $\mu(n)$.

2. Let $p\neq q$ be primes, $\zeta=e^{2\pi i/pq}$, and let $a_0,\dots,a_{pq-1}$ be non-negative integers with $\sum_k a_k\zeta^k=0$. Prove that $\sum_k a_k x^k$ is congruent modulo $x^{pq}-1$ to a combination with non-negative integer coefficients of the polynomials $x^r(1+x^q+\cdots+x^{(p-1)q})$ and $x^r(1+x^p+\cdots+x^{(q-1)p})$, that is, the roots form a union of regular p-gons and q-gons. (Hint: identify $\mathbb{Z}/pq$ with $\mathbb{Z}/p\times\mathbb{Z}/q$ and use $[\mathbb{Q}(\zeta):\mathbb{Q}]=(p-1)(q-1)$.) Deduce the Lam–Leung theorem for $n=pq$, and show that for $n=30$ the six numbers $e^{2\pi ik/5}$ $(k=1,2,3,4)$, $e^{\pi i/3}$, $e^{5\pi i/3}$ have zero sum although no subset of them is a regular polygon.

3. Let p and q be prime numbers with $p < q$. Suppose that in a convex polygon $P_1P_2 \dots P_{pq}$ all angles are equal and the side lengths are distinct positive integers. Prove that

$$P_1P_2 + P_2P_3 + \cdots + P_kP_{k+1} \;\geq\; \frac{k^3 + k}{2}$$

holds for every integer k with $1 \leq k \leq p$. (IMC 2018)

4. Let p be a prime number and let k be a positive integer. Suppose that the numbers $a_i = i^k + i$ for $i = 0, 1, \dots, p-1$ form a complete residue system modulo p. What is the set of possible remainders of a_2 upon division by p? (IMC 2023)

5. Let n be a positive integer and $a_1, \dots, a_n$ arbitrary integers. Suppose that a function $f\colon \mathbb{Z} \to \mathbb{R}$ satisfies $\sum_{i=1}^{n} f(k + a_i\ell) = 0$ whenever k and ℓ are integers and $\ell \neq 0$. Prove that $f = 0$. (IMC 2007)